# PROVISIONAL PATENT APPLICATION — DRAFT

## Method and Apparatus for Producing Discrete Quantized Changes in the Measured Weight of a Body by Driven Topological Reconfiguration of a Coherent Angular-Momentum Configuration

**Inventor:** Christopher F. Milam **Filing type:** U.S. Provisional Application under 35 U.S.C. §111(b)

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### CROSS-REFERENCE TO RELATED APPLICATIONS

This application is one of a coordinated series directed to applications of controlled angular-momentum coherence in condensed media. This application is companion to and distinct from applicant's earlier provisional directed to *continuous* weight modulation by coherent collective phase advance in an ordered medium (the "companion continuous-mode application"). The two applications recite complementary methods with distinct signatures: the companion continuous-mode application produces a smoothly varying support-force change as a function of applied drive power; the present application produces a **stair-step, discretely quantized** support-force change in fixed-magnitude events occurring at specific drive-parameter values corresponding to topological reconfiguration thresholds of a coherent angular-momentum configuration. Companion drafts also include transportation and lift systems; measurement instruments; energy storage and infrastructure; media and array architectures. Each companion incorporates this disclosure by reference upon filing.

### FIELD

Weight modulation and mass metrology by quantized force events. More particularly: methods and apparatus for producing a controlled, commanded, sign-selectable change in the measured weight of a body in the form of **discrete, fixed-magnitude force events** triggered at predetermined drive-parameter values by driving a coherent angular-momentum configuration across topological linking-number integer boundaries — equivalently, quantized additions or subtractions to the radius between the centers of extent of any two masses, occurring in integer-count events rather than as a continuous function.

### BACKGROUND

1. Prior art in weight modulation by angular-momentum effects is treated in the companion continuous-mode application (Wallace 1971; Laithwaite 1974; Alzofon 1981; Pais U.S. Pat. No. 10,144,532 B2). All named references address *continuous* modulation scaled with drive power or with rotation rate; none teaches a mechanism producing *discrete quantized events at threshold drive values.*
2. Prior art in topological quantization of angular-momentum states is well-established as pure physics but has not been reduced to weight-modulation practice:

- **Magnetic fluxon quantization** — the flux through a superconducting loop is quantized in units of $\Phi_0 = h/2e$ (Deaver–Fairbank 1961; London 1948, predicted). Established at metrological standard; the basis of the SQUID magnetometer and of the Josephson volt (2019 SI redefinition, $K_J = 2e/h = 483{,}597.848$ GHz/V; a commercial industrial receipt at ppb precision).
- **Vortex reconnection in fluid dynamics** — two adjacent vortex lines can locally exchange partners, changing mutual linking number by ±1; observed in classical fluids (Kleckner–Irvine 2013) and quantum fluids (Bewley–Paoletti–Sreenivasan 2008).
- **Skyrmion nucleation and annihilation** — topological soliton events in chiral magnets, individual events observed by real-time imaging (Romming et al. 2013). None of the above has been employed as an actuator producing a commanded support-force change in fixed-magnitude events on a laboratory scale, and none has been claimed as a weight-modulation apparatus.

3. What the art lacks — and this application supplies — is: (i) the **coherent angular-momentum configuration** whose driven topological reconfiguration events produce weight-modulation increments of fixed and measurable magnitude; (ii) the **drive geometry and control** that positions the configuration at the successive topological thresholds under a swept drive parameter, producing an integer-count sequence of events; and (iii) the **differential, artifact-rejecting measurement protocol** that isolates the discrete stair-step signature from any smooth background and from every recognized artifact.

**Applicant's published exhibit corpus (nonessential background).** Applicant maintains a public, commit-dated exhibit series presenting the underlying research program in didactic depth (the Normal Realism / POAMS exhibits, https://cfmilam.github.io/nr-poams-exhibits/; in particular *Counter-Rotating Planar Vortex Cores*, section 4). Those pages are applicant's own prior publications and are cited as nonessential background teaching only: no material essential to enablement, written description, or the claims is incorporated from them, and this specification is complete and self-contained.

## SUMMARY

A method of altering the measured weight of a body in discrete quantized events comprises: providing a body comprising a coherent angular-momentum configuration having a topological linking-number invariant; applying a static reference orientation that establishes a signed relationship between said configuration and the body's natural-orbit angular-momentum vector; applying a swept drive parameter (drive power, drive frequency, imposed field, mutual inductance, or geometric compression) that carries said linking-number invariant across successive integer boundaries; sensing the resulting stair-step sequence of fixed-magnitude support-force events with a differential, artifact-rejecting protocol; and selecting the sign of said events by selecting the reference orientation.

Apparatus comprises the body, a drive source arranged to sweep the drive parameter across predetermined ranges spanning multiple linking-number integer boundaries, a reference- orientation source, a force sensor of resolution sufficient to detect a per-event force change, and a controller executing the differential protocol and enumerating events.

The per-event force change is a fixed quantum whose magnitude is set by the coherent domain size and drive turn rate; total accumulated force change is the sum over triggered events. Sign is selected by reference-orientation sense; magnitude per event is a design and material choice, calibrated at manufacture. Modules embodying the method provide calibrated, quantized weight-trim increments — a discrete-count actuator rather than a continuous-analog trim — with applications in precision-force metrology, digital force control, and calibrated payload adjustment.

### BRIEF DESCRIPTION OF THE DRAWINGS

- FIG. 1 — System block diagram: coherent-AM device on force sensor; drive source with sweep controller; reference-orientation source; detector chain; event-enumerating controller; data log.
- FIG. 2 — Device detail (reference embodiment): counter-wound superconducting coil pair in stacked-not-merged geometry, mutual-inductance axis vertical, insulating support between coils, cryostat and thermal management.
- FIG. 3 — Drive-parameter sweep timing: monotonic ramp of drive current or mutual inductance, with expected event positions marked at integer-Lk thresholds.
- FIG. 4 — Force sensor trace (prophetic): stair-step response as drive parameter crosses each integer boundary, with per-event $\Delta g$ step and quiescent plateau between events.
- FIG. 5 — Sign-selection map: reference-orientation sense (aligned/anti-aligned with natural-orbit vector) vs. event-sign polarity.
- FIG. 6 — Alternative embodiments: (a) superfluid vortex-ring pair in Helium-II container; (b) skyrmion-lattice thin film under magnetic-field sweep; (c) Josephson-junction array under bias-current sweep.
- FIG. 7 — Quantized-trim module (OEM package): sealed device with electrical interface accepting integer event-count command and reporting achieved cumulative force offset.
- FIG. 8 — Array embodiment: N devices in parallel with staggered threshold values for fine-grained cumulative control.

### DEFINITIONS (the applicant acts as his own lexicographer; these definitions govern the claims)

1. **Coherent angular-momentum configuration** — a physical structure in which a macroscopically observable population of internal angular-momentum carriers maintains collective phase coherence such that a well-defined net vector angular momentum characterizes the population as a whole; the extant labels include: persistent supercurrent loop, aligned magnetization domain, vortex ring, skyrmion, Josephson-array phase-locked state.

2. **Topological linking-number invariant (Lk)** — the signed integer linking number of a closed circulation with itself or with a second circulation in its neighborhood (Călugăreanu 1959; the flux quantum h/2e of London 1948 is one measured form; the winding number of a vortex ring is another). Operationally: the integer count of completed wraps of the coherent angular-momentum configuration as read by an interferometric or flux-linkage measurement.
3. **Reference orientation** — the direction in space defining the sign convention for coupling between the configuration's angular momentum and the local natural-orbit angular-momentum vector; in laboratory practice, an axis parallel to Earth's rotation axis at the equator, or the natural-orbit axis at the device's latitude, or any axis under the operator's control aligned or anti-aligned to said natural-orbit vector.
4. **Swept drive parameter** — the operator-controlled physical quantity whose monotonic variation carries Lk across successive integer boundaries. Depending on embodiment, the drive parameter is (a) the current through a superconducting coil pair, (b) the applied magnetic-field magnitude across a skyrmion lattice, (c) the bias current across a Josephson junction array, (d) the mutual-inductance separation between two coherent coils, or (e) the compression parameter of a vortex-ring pair.
5. **Integer boundary / linking-number threshold** — a specific value of the drive parameter at which Lk of the coherent configuration transitions between two adjacent integer values. Operationally: the drive-parameter value at which an event fires in the differential force protocol.
6. **Discrete quantized force event / stair-step transition** — an isolated, single-step change in the sensed support force occurring at an integer boundary of the drive parameter, of fixed magnitude characteristic of the device and materials, and separated from adjacent events by quiescent drive-parameter intervals in which the force reading is stable.
7. **Coherent-domain size (N_local)** — the count of angular-momentum carriers in the phase-coherent domain that reorganizes at a single integer-boundary crossing. Set by material choice, apparatus geometry, and drive amplitude; measured operationally by the per-event force change divided by the single-carrier force contribution at the device's drive turn rate.
8. **Drive turn rate (ν_drive)** — the frequency (hertz) at which the coherent configuration's collective phase advances under the applied drive; for a supercurrent coil the AC drive frequency; for a magnetic-field sweep, the effective precession frequency at the imposed field.
9. **Per-event force quantum (ΔF_event)** — the magnitude of the support-force change at a single integer-boundary crossing, operationally equal to (per the applicant's teaching framework, non-limiting) approximately N_local · h · ν_drive divided by a device- specific effective length. Measured value governs; the teaching formula is not claimed.
10. **Support force / measured weight** — the force read by a sensor supporting the body (newton), under the differential protocol of this disclosure.
11. **Differential event-enumerating protocol** — the measurement protocol comprising: monotonic ramp of the drive parameter through the range expected to contain multiple integer boundaries; synchronous logging of the support-force channel; identification of stair-step transitions by threshold-crossing of the force-derivative; enumeration and cataloguing of events with drive-

parameter position, sign, and magnitude; validation gates that the events (a) recur at reproducible drive-parameter values on repeat, (b) reverse sign under reference-orientation reversal, (c) do not fire under detuned drive off the expected threshold, (d) are absent for a non-coherent reference body under identical protocol.

12. **Ordered magnetic medium** — a medium exhibiting spontaneous or field-induced long-range alignment of internal circulations, whose macroscopic magnetization provides the net vector angular momentum required by Definition 1. Reference class: iron garnets (YIG and substituted variants), skyrmion-hosting materials (MnSi, FeGe, $Cu_2OSeO_3$), doped corundum.
13. **Superconducting coil pair configuration** — two independently-driven coherent supercurrent loops of specified mutual inductance arranged coaxially (aligned axes) with an insulating separation preventing physical merger but permitting mutual flux linkage; the pair sustains an Lk invariant that is a function of the two loop currents and their mutual inductance.
14. **Counter-wound (opposite-sense) geometry** — the coil-pair configuration in which the two loops carry supercurrents in opposite azimuthal senses about their common axis; the net axial angular momentum sums (magnitudes add along the coaxial line) while the total mutual linking with any external body can be tuned via mutual inductance without merging the loops.

## THEORETICAL FOUNDATION (teaching; the claims are mechanism-free)

**T1. Units and rates.** Energy is turn rate: $E = h\nu$, with h the action of one completed turn of internal phase. Mass is trapped rate: $m = E/c^2 = h\nu/c^2$. These identities are standard algebra on measured constants, consistent with the SI's own realization chain (the 1967 second: 9,192,631,770 turns of Cs-133 hyperfine; the 2019 volt: $2e/h = 483{,}597.848$ GHz/V). The teaching of the companion continuous-mode application §T1 is incorporated by reference; no new units teaching is asserted here.

**T2. Weight as constraint force.** As taught by the companion continuous-mode application §T2 (incorporated by reference): a body's measured weight is the constraint force required to hold it off its natural orbit at the sensed radius; that natural orbit is a function of the body's angular-momentum state; therefore commanded modification of the body's angular-momentum state produces commanded change in weight. Applicable measured receipts (GPS clock offset, Mercury perihelion advance, orbital-time-dilation factor of three) verified at metrological standard.

**T3. Topological quantization of coherent angular-momentum configurations.** The linking- number invariant Lk of a coherent angular-momentum configuration takes integer values. Below is a partial list of established industrial and metrological receipts for this proposition; each is standard physics and none is claimed by this application:

- Magnetic flux through a superconducting loop is quantized in $\Phi_0 = h/2e$; each SQUID magnetometer measures individual $\Phi_0$-events by transitions of the loop's linking number.
- Josephson junctions phase-lock at integer multiples of $2\pi$ radians of phase difference; the Shapiro steps in the current-voltage characteristic of a driven junction are integer-indexed and are the basis

of the metrological volt.

- Vortex rings in classical fluids carry integer circulation quanta; two-ring systems undergo reconnection events changing mutual linking number by ±1 at critical compressions.
- Skyrmion nucleation in chiral magnets adds skyrmion count by ±1 at critical field-magnitude thresholds; observed by real-time Lorentz TEM. Common to all: the linking or winding number is a topological invariant that transitions between adjacent integer values in fixed discrete steps at critical values of a control parameter. **The present application's mechanism is a driven traversal of such transitions, applied to produce corresponding transitions in the constraint-force balance of a body.**

**T4. Coupling of coherent angular-momentum to natural-orbit vector.** As taught by the companion continuous-mode application §T5 and by the linking-decomposition (Călugăreanu Lk = Tw + Wr), the sign of the coupling between a driven coherent angular-momentum configuration and the body's natural-orbit angular-momentum vector is set by their relative orientation; co-aligned configurations contract the natural orbit and increase measured weight; anti-aligned configurations expand the natural orbit and decrease measured weight; perpendicular configurations produce no first-order effect. This teaching is incorporated by reference from the companion; no new sign teaching is asserted here.

**T5. Per-event force quantum.** At an integer-boundary crossing of Lk, the coherent angular-momentum configuration undergoes a topological reconfiguration event; a coherent domain of size N_local (the count of angular-momentum carriers phase-locked in the transitioning region) simultaneously reorganizes; the total angular-momentum increment delivered in the event is approximately N_local · ħ; by the coupling of §T4, the corresponding support-force change per event is of order: ΔF_event ≈ (N_local · ħ · ν_drive) / ℓ_eff, where ν_drive is the drive turn rate and ℓ_eff is a device-specific effective moment arm. For a superconducting coil pair of the reference embodiment (E1 below), N_local ranges from $10^{6}$ (fluxon-scale local reorganization) to $10^{20}$ (full-condensate reorganization), producing per-event force quanta from $10^{-18}$ N (sensitive-calorimetry class) to $10^{-2}$ N (macroscopic-balance class), depending on drive amplitude and coherent-domain size.

**T6. Energy accounting (operability).** The apparatus is an energy converter. The energy of each topological reconfiguration event is drawn from the drive; the drive is electromagnetic and its energy is metered at the input port. Cumulative energy input over a sweep containing N events is bounded below by N · N_local · h · ν_drive plus any adiabatic losses and cavity losses. No step creates energy; drive termination and Lk-reset return the device to its baseline state with the associated body returning to its undriven weight. The device claims no perpetual motion and permits none.

**T7. Sign-selectability and reproducibility.** The sign of each event is determined by the reference orientation of §T4; reversing the reference orientation (via applied field polarity, geometric rotation of the device, or reversal of one of the paired coil currents) reverses the sign of the events. The drive-parameter positions of the successive integer boundaries are reproducible under fixed material and geometric parameters; a recorded threshold catalog specific to a given device permits open-loop commanded event firing without contemporaneous force measurement, matching the calibration-based open-loop operation taught by the companion continuous-mode application §E5 (variant).

## DETAILED DESCRIPTION OF EMBODIMENTS

**E1 — Reference embodiment: counter-wound superconducting coil pair with quantized-event force actuation.** Two coaxial superconducting coils, each wound from Nb-Ti or NbSn wire, mounted on a common insulating former (G-10 or PEEK) in stacked-not-merged geometry (axial separation 0.5–5.0 cm; coil radii 1–10 cm; turns per coil 100–1000). The two coils are wound in opposite azimuthal senses (Definition 14). Each coil is driven independently through persistent-mode switches from a shared programmable current source with independent per-coil current control (reference: 0–100 A, controlled to ±10 mA precision, sweepable at 0.01–1 A/s). The pair rests on a low-thermal-loss support at the drive-field antinode of the containment vessel; the assembly is enclosed in a liquid-helium cryostat maintaining < 4.2 K over the coil volume. A precision force sensor (analytical balance, readability ≤ 0.1 μg; or piezoelectric load cell of equivalent resolution) supports the assembly. The reference-orientation vector is established by the physical orientation of the coil-pair axis: mounted vertically at any latitude, the axis is nominally parallel to the local zenith; the operator rotates the assembly to establish the desired sign-convention alignment with respect to the natural-orbit angular-momentum vector (nominally polar at the equator; latitude-dependent).

Controller operating sequence (differential, event-enumerating; the programmed protocol the controller executes): (1) baseline with both coils in normal state at cryogenic temperature, no drive current, force sensor zeroed; (2) coil A ramped to reference current I_A; persistent-mode switch closed; baseline recorded; (3) coil B ramped from zero through the expected sweep range at controlled rate (0.01–1 A/s), while force sensor is logged at 1 kHz bandwidth; (4) sweep segmented into windows; stair-step transitions identified by force-derivative threshold-crossing detector; events catalogued with (drive-parameter value, sign, magnitude); (5) coil B reset to zero; sweep repeated to establish reproducibility of threshold positions (validity gate A1); (6) coil A polarity reversed; entire sweep repeated; each event must reverse sign (validity gate A2); (7) coil A reset to zero; sweep of coil B repeated in absence of coil A; expected to produce no events (validity gate A3 — coupling requires paired configuration); (8) reference body of same mass but non-superconducting composition substituted; full protocol repeated; expected to produce no events (validity gate A4 — material specificity).

Validity gates (programmed before operation; the controller reports events only when all pass): A1 threshold-position reproducibility ≤ 1% RMS on repeat sweep; A2 sign reversal under coil-A polarity reversal; A3 no events under single-coil drive; A4 material specificity vs. non-coherent reference body; each declared at ≥3σ on the force channel. Auxiliary diagnostics: A5 event-firing time-signature (per-event risetime; the topological transition should present as a step change on the timescale of the coherent-domain reorganization dynamics, faster than the drive-parameter sweep); A6 drive-power balance check (each event must draw the expected N_local · h · ν_drive energy from the drive input, verified at the current-source metering port).

**E2 — Full quantized weight offset: cumulative-event station-keeping.** Operation, stated natively: the drive sweeps the drive parameter through successive integer boundaries, firing events in sequence; each event contributes a fixed force quantum to the cumulative weight change; the operator halts the sweep at the desired cumulative offset count; the body then rests at a support force reduced (or increased) by the summed event quanta from the undriven baseline. Apparatus: as E1, with: extended sweep range (drive parameter ranged over $10^2$–$10^5$ event integer boundaries); event-counting register in the controller; interlock to halt sweep at commanded count. The station-keeping regime is maintained by the persistent-mode current lock — no active drive is required between events during the sustained-offset phase.

**E3 — Alternative embodiment: skyrmion-lattice thin-film with magnetic-field sweep drive parameter.** Thin-film sample of chiral magnet (MnSi, FeGe, or $Cu_2OSeO_3$; thickness 50–500 nm; area 1–100 mm$^2$) mounted on a piezoelectric force sensor in a cryostat operating at 20–200 K (temperature range and film composition selected to place the sample in the skyrmion-phase region of its phase diagram). A perpendicular applied magnetic field is swept through the skyrmion-phase field range (typical 0.1–1 T for standard chiral magnets; specific values material-dependent) with monotonic ramp. Each nucleation or annihilation of a skyrmion is one topological event; the imposed magnetic field is the drive parameter; the event-count corresponds to the skyrmion-population count. The force sensor detects per-skyrmion force quanta; validity protocol as in E1 with skyrmion-count microscopy (Lorentz TEM or MFM) confirming event enumeration.

**E4 — Alternative embodiment: Josephson-junction array under bias-current sweep.** A parallel array of Josephson junctions on a common superconducting substrate, mounted on a precision force sensor in a helium cryostat. Bias current is swept through the array under controlled ramp; Shapiro-step transitions are the events; each transition corresponds to an integer-Lk change of the array's collective phase. Force per event enumerated as in E1.

**E5 — Materials envelope.** Any coherent angular-momentum configuration supporting well-defined Lk transitions at threshold drive values, and possessing sufficient coherence to maintain a coherent domain of size N_local sufficient to produce detectable per-event force quanta. Preferred classes: superconducting coil pairs (Type I or Type II superconductors; the coils may independently be normal-metal for reduced N_local at the cost of dissipation); skyrmion- lattice thin films of chiral magnets; Josephson-junction arrays; superfluid Helium-II vortex-ring pair configurations; persistent-current loops in coherent metals at sub-Kelvin temperatures. Engineered and composite structured media (metamaterials) — layered, doped, strained, or geometrically patterned — are expressly within the envelope wherever they provide the requisite coherent-AM configuration and Lk transitions.

**E6 — Quantized weight-trim actuator module (OEM).** A sealed module comprising: coherent-AM device, drive electronics, reference-orientation source (mechanical mount or applied bias field), force/acceleration sensor, integrated event-counting controller with electrical interface accepting an integer event-count setpoint and reporting achieved cumulative force offset, lock state, and health. Mountable to payloads, instruments, platforms, and vehicles. The module is the unit article of commerce; a plurality of modules with staggered threshold values provides fine-grained cumulative force control.

**E7 — Trim-class use methods.** Methods of operating E6 modules to: apply a commanded integer count of quantized force events (positive or negative sign); dither a cumulative offset by fire-and-reset event pairs; balance a platform by firing sign-selectable events across a plurality of modules; provide calibrated force injection for precision-force metrology (each event is a traceable force quantum from device calibration).

## PROPHETIC EXAMPLES (paper examples; no data represented as performed)

**P1.** A counter-wound Nb-Ti coil pair per E1 (coil radii 5 cm, 500 turns per coil, axial separation 1 cm, cryostat at 2 K) with coil A at persistent-mode 50 A, coil B swept 0–50 A at 0.1 A/s, is expected on the framework's scaling to display $10^2$–$10^4$ discrete force events during the sweep, each of magnitude $10^{-15}$ to $10^{-11}$ N per event, with threshold reproducibility at the ±1% RMS gate, distinguishable from the balance's readability floor by co-summation over $10^3$ successive events. Sign of each event reverses under coil A polarity reversal (validity gate A2). Cumulative offset over the full sweep is expected to be in the $10^{-12}$–$10^{-8}$ N range, measurable on a laboratory analytical balance of readability $10^{-10}$ N (sub-microgram).

**P2.** A larger-scale coil pair per E1 (coil radii 20 cm, 1000 turns per coil, axial separation 5 cm, larger cryostat) driven at 100 A currents is expected to display per-event force quanta in the $10^{-10}$ to $10^{-6}$ N range, individually detectable on a precision force sensor (piezoelectric or capacitive-diaphragm load cell of readability $10^{-12}$ N) during a slow (0.01 A/s) sweep. Cumulative offset over a full sweep in the $10^{-8}$ to $10^{-4}$ N range.

**P3.** A skyrmion-lattice thin-film per E3 (MnSi film, 200 nm thickness, 5 $mm^2$ area, operating at 25 K, magnetic field swept 0.2–0.4 T) is expected to display $10^3$–$10^5$ skyrmion-nucleation and annihilation events across the sweep, each corresponding to a per-skyrmion force quantum of order $10^{-18}$ N per event (individual detection near noise floor; cumulative $10^{-13}$ N over the sweep, measurable by summation). The nature of this alternative embodiment provides independent cross-verification of the stair-step signature via correlated microscopy (real-time Lorentz TEM confirms skyrmion count).

**P4.** A quantized weight-trim module per E6 with calibrated event count and per-event force quantum (calibration by direct force measurement at manufacture) is expected to produce commanded cumulative force offsets to within one event-count of the setpoint, open-loop, without contemporaneous force feedback; the module's specification would guarantee, e.g., "1000 events per activation cycle, $\pm 10^{-10}$ N per event, cumulative $\pm 10^{-7}$ N."

## CLAIMS

1. A method of altering the measured weight of a body in discrete quantized events, comprising: providing a body comprising a coherent angular-momentum configuration having a topological linking-number invariant Lk; establishing a reference orientation of said configuration relative to said body's natural-orbit angular-momentum vector; sweeping a drive parameter monotonically through a range containing multiple integer boundaries of said linking-number invariant; sensing a resulting stair-step sequence of discrete support-force events, each event corresponding to a transition of said linking-number invariant across an integer boundary; and selecting the sign of said events by selecting the sense of said reference orientation.
2. The method of claim 1, wherein said sensing comprises a differential event-enumerating protocol comprising: identification of stair-step transitions by force-derivative threshold-crossing; enumeration of events with drive-parameter position, sign, and magnitude; and validation gates that events (a) recur at reproducible drive-parameter values on repeat sweep, (b) reverse sign under reference-orientation reversal, (c) do not fire in absence of the coherent configuration, and (d) are absent for a non-coherent reference body under identical protocol; each declared at $\geq 3\sigma$ on the force channel.
3. The method of claim 1, wherein the coherent angular-momentum configuration comprises a counter-wound coil pair of superconducting wire arranged coaxially in stacked-not-merged geometry.
4. The method of claim 1, wherein the coherent angular-momentum configuration comprises a skyrmion lattice in a chiral magnetic thin film.
5. The method of claim 1, wherein the coherent angular-momentum configuration comprises a parallel array of Josephson junctions on a superconducting substrate.
6. The method of claim 1, wherein the coherent angular-momentum configuration comprises a pair of vortex rings in a superfluid.
7. The method of claim 1, wherein the drive parameter is the current in one loop of a counter-wound coil pair, and said sweeping comprises monotonic ramp of said current from zero to a value spanning multiple integer boundaries of the pair's mutual linking-number invariant.
8. The method of claim 1, wherein the drive parameter is the applied magnetic field on a skyrmion-lattice thin film, and said sweeping comprises monotonic ramp of said field through the skyrmion-phase field range.
9. The method of claim 1, wherein the drive parameter is the bias current on a Josephson-junction array, and said sweeping comprises monotonic ramp of said current through multiple Shapiro-step transitions.

10. The method of claim 1, further comprising halting said sweeping at a commanded integer event-count, whereby a cumulative support-force offset equal to the sum of the per-event force quanta over the fired events is established.
11. The method of claim 10, wherein said cumulative offset is maintained by persistent- mode current lock without active drive during the sustained-offset phase.
12. The method of claim 1, wherein said reference orientation is established by physical mounting of said coherent angular-momentum configuration with its principal axis parallel or anti-parallel to a chosen direction of said natural-orbit angular-momentum vector.
13. The method of claim 1, further comprising recording a threshold catalog of the drive- parameter values at which events fire for a given device, and subsequently commanding integer event counts open-loop by drive-parameter setpoint without contemporaneous force measurement.
14. Apparatus for producing discrete quantized changes in the measured weight of a body, comprising: a body comprising a coherent angular-momentum configuration having a topological linking-number invariant; a reference-orientation source establishing said body's principal axis relative to a natural-orbit angular-momentum vector; a drive source arranged to sweep a drive parameter monotonically through a range containing multiple integer boundaries of said linking-number invariant; a force sensor arranged to measure the support force of at least said body; and an event-enumerating controller configured to execute a differential event-enumerating protocol and to catalog fired events by drive-parameter position, sign, and magnitude.
15. The apparatus of claim 14, wherein said coherent angular-momentum configuration comprises a counter-wound superconducting coil pair mounted in a cryostat.
16. The apparatus of claim 14, wherein said coherent angular-momentum configuration comprises a chiral-magnet thin film mounted for perpendicular magnetic-field drive.
17. The apparatus of claim 14, wherein said coherent angular-momentum configuration comprises a Josephson-junction array on a superconducting substrate.
18. The apparatus of claim 14, further comprising an event-counting register in said controller, and an interlock configured to halt said sweep at a commanded integer event count.
19. The apparatus of claim 15, further comprising a persistent-mode switch on each coil of the pair, arranged to lock each loop's supercurrent independently of the others following completion of the drive parameter sweep.
20. The apparatus of claim 14, further comprising a mechanical mount arranged to rotate the entire coherent angular-momentum configuration about at least one axis, thereby establishing selectable reference-orientation sense relative to said natural-orbit angular-momentum vector.
21. A quantized weight-trim actuator module comprising the apparatus of claim 14 in a sealed housing with an electrical interface configured to accept an integer event- count command and to report achieved cumulative support-force offset, event-count register, lock state, and health.

22. A method of trimming the measured weight of a payload element, comprising mounting thereto a module per claim 21 and commanding an integer event count to produce a selected cumulative support-force offset of said payload.
23. An array apparatus comprising a plurality of modules per claim 21, each module having a different characteristic per-event force quantum, arranged and commanded to produce a fine-grained cumulative support-force offset across said array.
24. The method of claim 1, wherein all mechanical work performed in raising said body during operation is supplied by the said drive, the method comprising terminating the drive and resetting said linking-number invariant to baseline, thereby restoring the undriven weight.
25. The method of claim 1, wherein said measuring is performed with the body and its drive assembly in a sealed enclosure at reduced pressure, whereby convective and radiometric artifacts are excluded.
26. The method of claim 1, wherein said coherent angular-momentum configuration comprises an engineered or composite structured medium (a metamaterial), natural or synthetic, providing said topological linking-number invariant and supporting said transitions.
27. Apparatus for producing a commanded discrete-quantized weight offset of a mass, comprising: a mass comprising a coherent angular-momentum configuration having a topological linking-number invariant; a reference-orientation source arranged in a selected sense relative to the mass's natural-orbit angular-momentum vector; a drive source arranged to sweep a drive parameter through a predetermined range containing a selected number of integer boundaries of said linking-number invariant; and a controller storing a threshold catalog and configured to command drive-parameter sweep in accordance therewith to fire a commanded number of quantized events, thereby establishing a commanded cumulative support-force offset of said mass.

## ABSTRACT

A body containing a coherent angular-momentum configuration with a topological linking- number invariant — exemplified by a counter-wound superconducting coil pair in stacked- not-merged geometry — is placed on a force sensor and driven by a monotonically swept drive parameter (loop current, magnetic field, or bias current, depending on embodiment) through a range containing multiple integer boundaries of said linking-number invariant. Each crossing of an integer boundary fires a discrete quantized force event of fixed magnitude on the body's support force; the sign of the events is selected by the reference orientation of the configuration relative to the body's natural-orbit angular- momentum vector; the total cumulative offset is the sum over triggered events. Apparatus, sweep protocols, alternative embodiments (skyrmion-lattice thin films, Josephson-junction arrays, superfluid vortex-ring pairs), sealed OEM modules with electrical integer-count interface, and prophetic performance estimates are disclosed, together with a differential event-enumerating measurement protocol and energy accounting establishing operability. Distinct in signature and mechanism from the applicant's continuous-

mode weight modulation companion application, the present application discloses a discrete-count actuator whose per-event force quantum is a calibrated device parameter and whose cumulative offset is commanded by an integer event-count setpoint.

**Fig. 1 — System block diagram (100)**

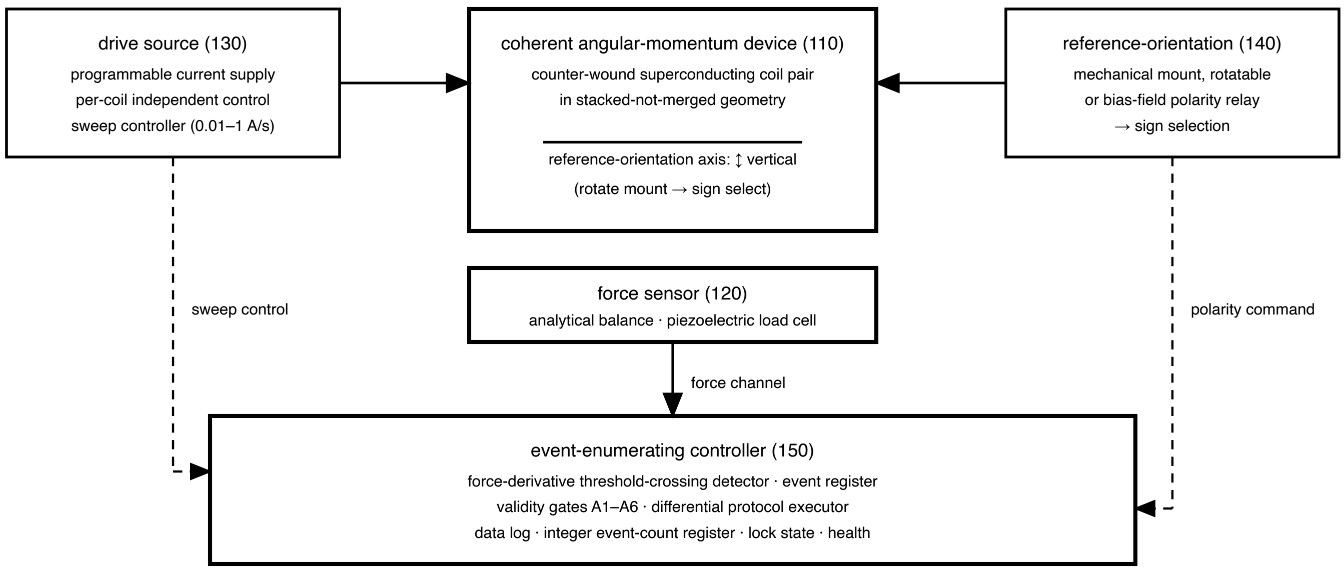
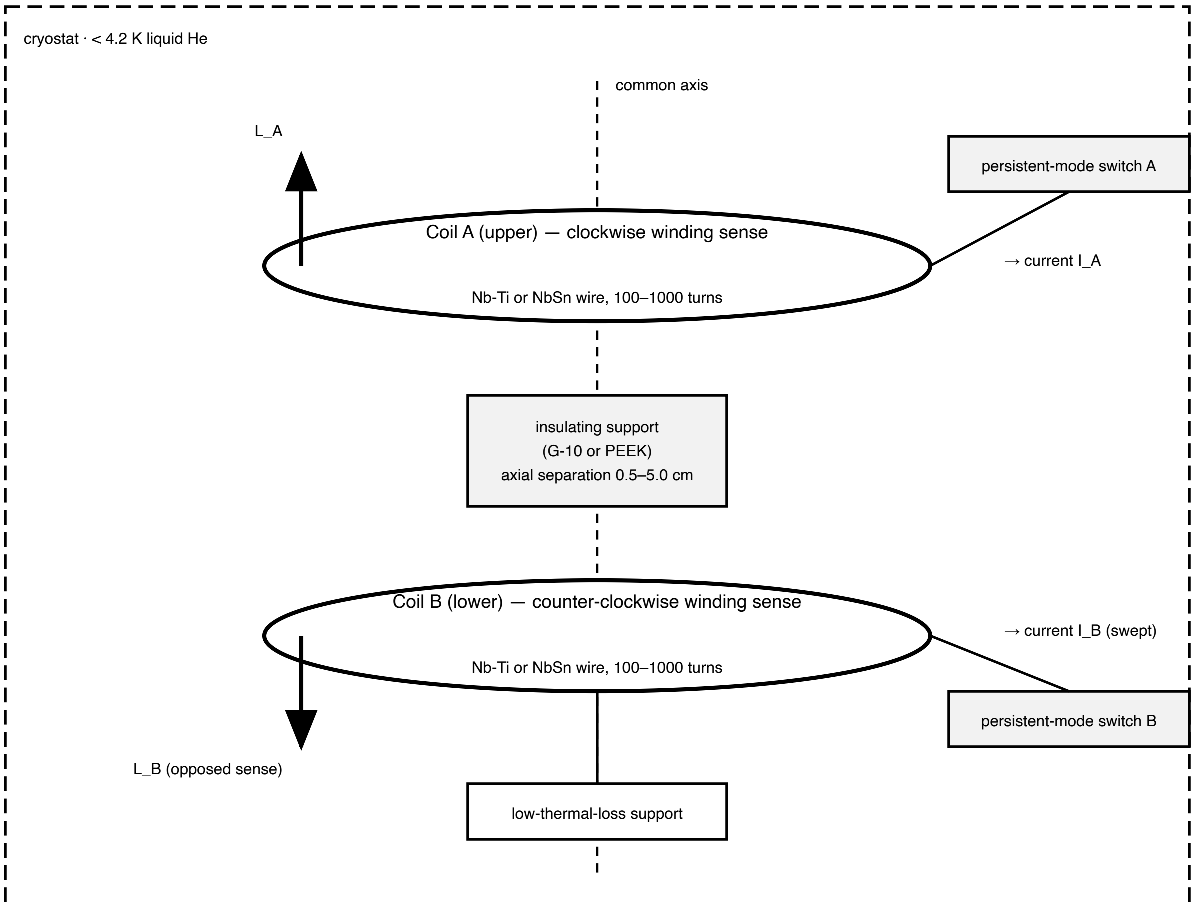
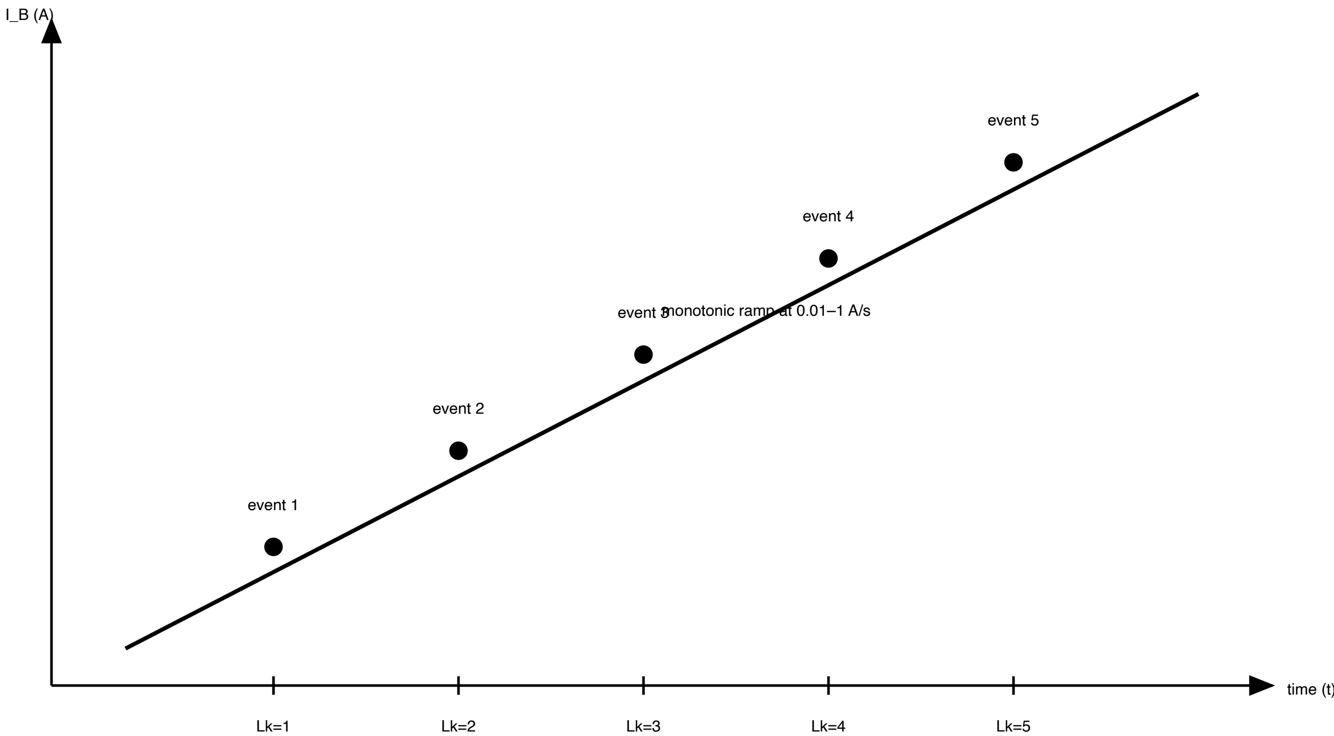


**Fig. 2 — Counter-wound superconducting coil pair, stacked-not-merged (200)**



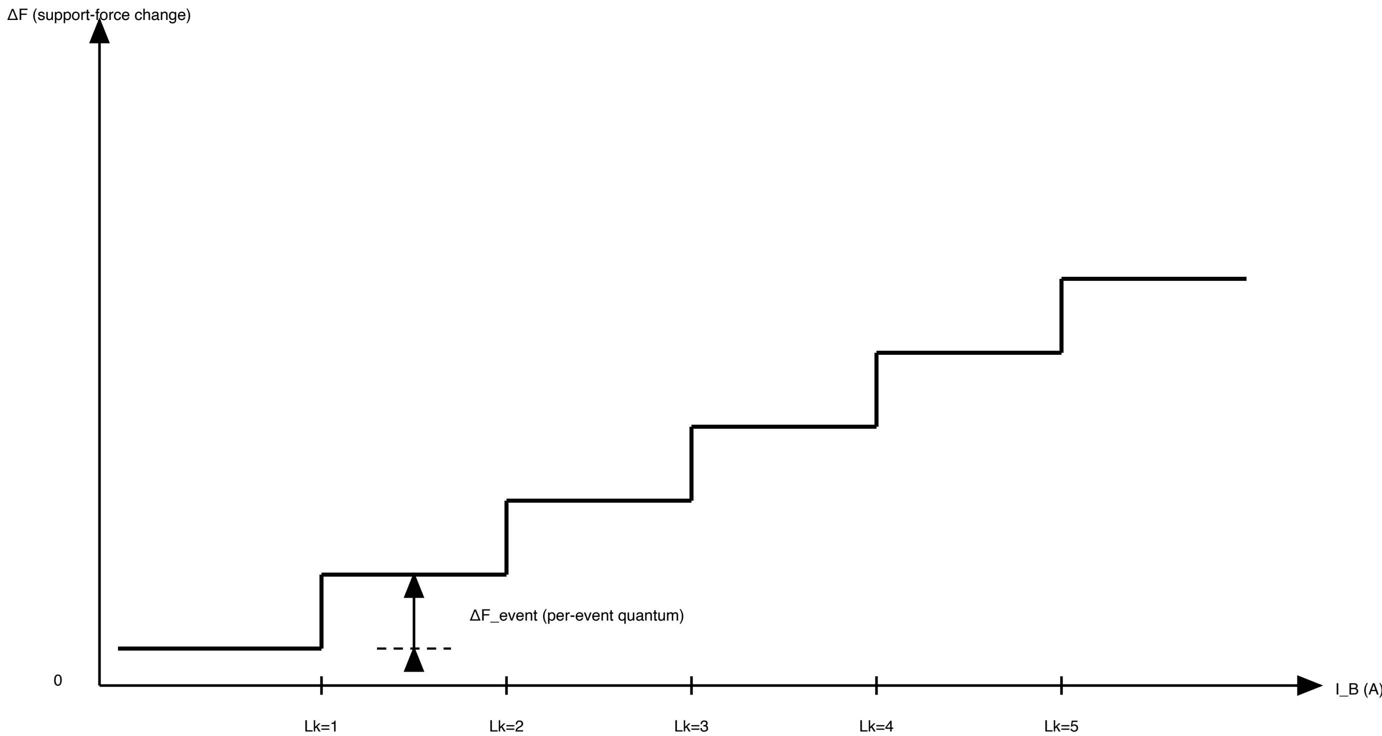

mutual linking-number Lk swept by ramp of I_B at fixed I_A

**Fig. 3 — Drive-parameter sweep timing (300)**



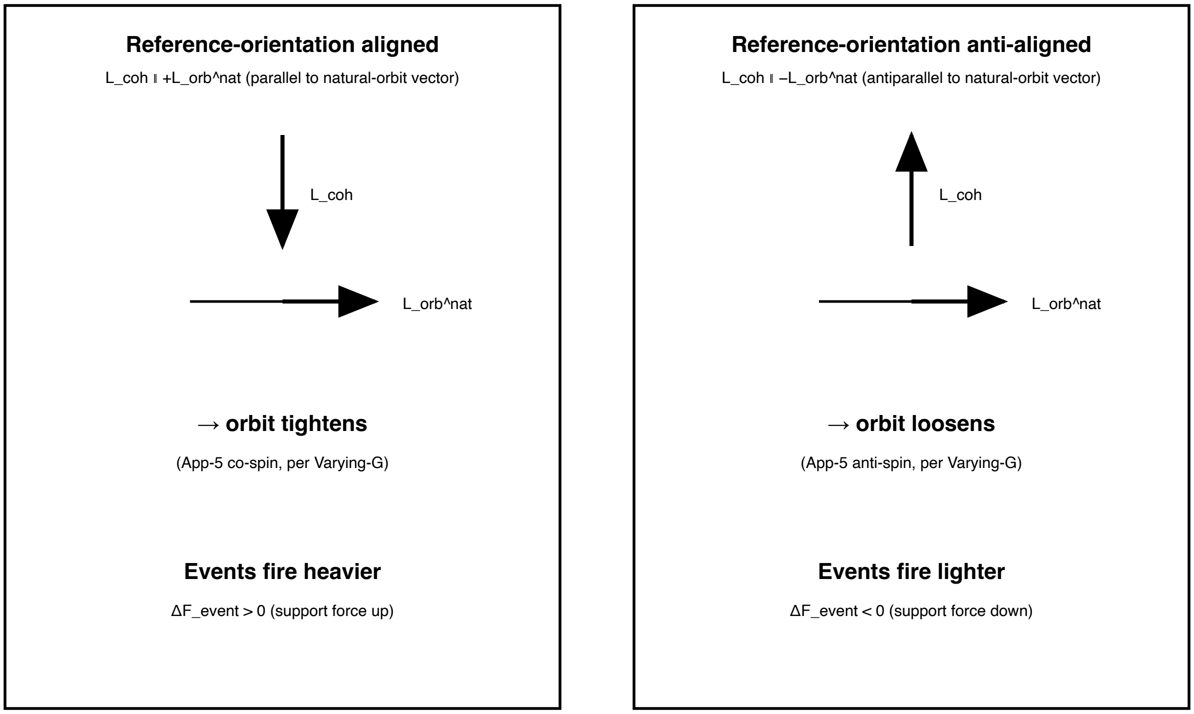

Each integer-Lk boundary crossing fires one discrete event.
Sweep rate and range chosen to span the expected event count.

**Fig. 4 — Force sensor trace (prophetic): stair-step response (400)**




Between crossings, force reading is stable (quiescent plateau).
Sign reversal under reference-orientation reversal gives inverted staircase.

**Fig. 5 — Sign-selection map (500)**




Sign selection: reverse the reference-orientation by rotating the device 180° about a horizontal axis.

Fig. 6 — Alternative embodiments (600)

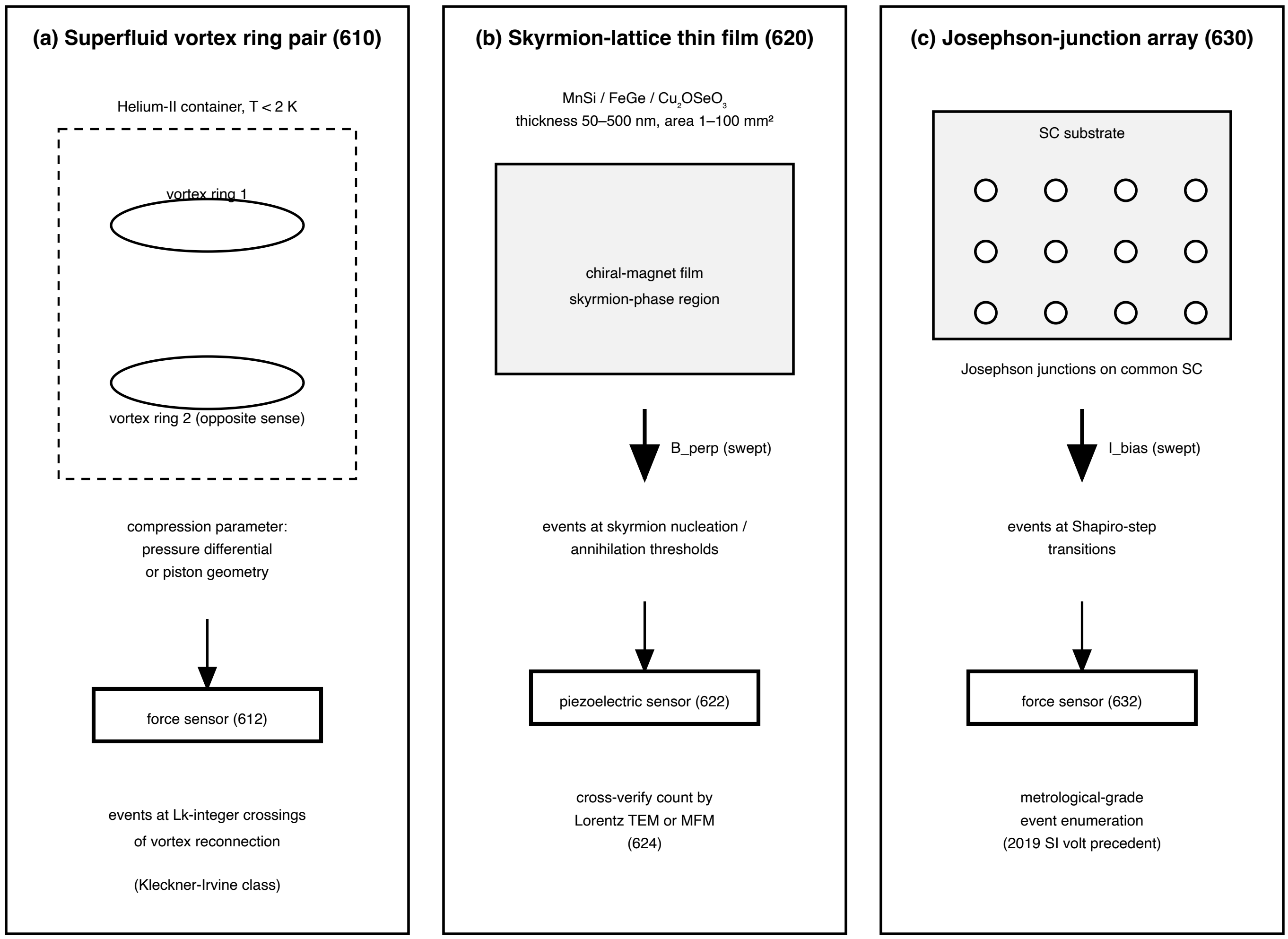
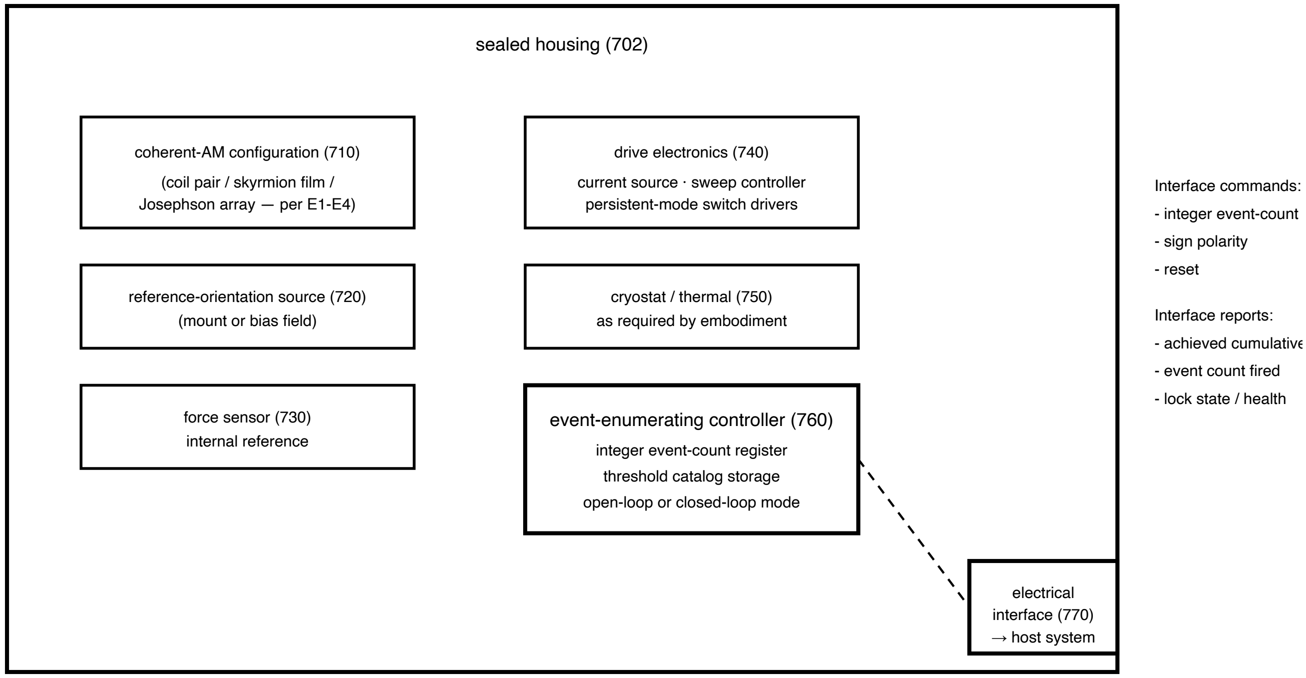
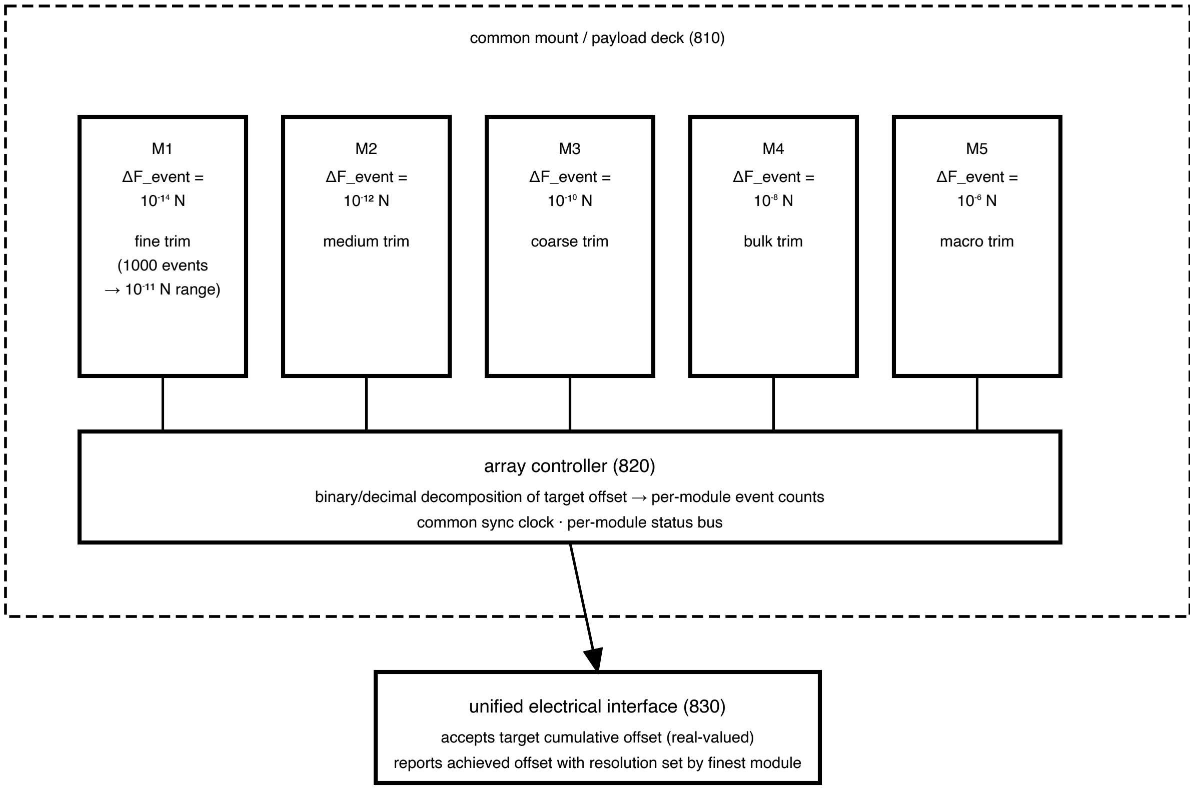


**Fig. 7 — Quantized weight-trim actuator module — OEM (700)**




Module is the unit article of commerce; the interface accepts a discrete-count setpoint
and reports achieved cumulative support-force offset, without requiring host-side force measurement.

**Fig. 8 — Array of quantized-trim modules with staggered per-event quanta (800)**




Fine-grained cumulative control by summation over modules of staggered per-event quanta;
the array acts as a digital force-offset actuator with resolution = finest module's ΔF_event.